Case Study : Bank Marketing

Predicting Customer Subscription to Term Deposits in Bank Marketing Campaigns

Dataset: ***bank.csv***

1. Introduction

Banks conduct marketing campaigns to promote financial products such as term deposits, credit cards, and loans. These campaigns often involve large operational costs, especially when executed through outbound channels like telephonic outreach. To improve efficiency and return on investment, banks need to **identify which customers are most likely to respond positively to the campaign** — in this case, whether a customer will **subscribe to a term deposit** after being contacted.

This case study uses the **Bank Marketing Dataset**, which contains customer demographic, economic, and campaign-related attributes. The goal is to build predictive models that estimate the likelihood of a customer subscribing to a term deposit — effectively turning raw data into actionable insights that can inform marketing strategies and customer targeting.

This problem is inherently a **binary classification task**, where the target variable indicates whether a customer responded positively (yes) or negatively (no) to a marketing campaign. The emphasis in this case study is on **industry-aligned modeling practices, metric selection**, and **deployment considerations** that would be applicable in real financial services environments.

2. Business Context

Marketing campaigns, particularly those executed via outbound contact (e.g., phone calls), involve significant time and expenditure. Call center resources, customer acquisition costs, and opportunity costs must all be balanced against the revenue generated when a customer subscribes to a term deposit.

By using data-driven prediction models, the bank can:

- Allocate resources more efficiently by targeting high-likelihood responders
- Reduce cost per subscription
- Improve customer experience by avoiding unnecessary contacts
- Increase overall campaign effectiveness

The business impact is measured not just by predictive accuracy, but by the **efficiency of the campaign in converting prospects into term deposit holders**.

3. Why Classification Is Required

This is a **binary classification problem** because:

- The target outcome has **two discrete categories**: subscribe (yes) or not subscribe (no).
- The objective is not to estimate a continuous value; rather, it is to assign a **probability and decision** of term deposit subscription.

Classification models allow us to:

- Output **probability scores** that represent the likelihood of subscription
- Compare customers on a risk/reward scale
- Tune **decision thresholds** based on business priorities (e.g., maximizing recall vs precision)
- Evaluate performance using appropriate metrics such as **ROC-AUC, Precision, Recall, F1-score**, and business utility curves

This modeling approach drives business decisions such as which customers to call, when to call, and which channels to prioritize.

4. Problem Definition

The case study focuses on developing, evaluating, and interpreting classification models that predict whether a customer will subscribe to a term deposit. Key elements include:

- **Prediction Target**: Binary subscription outcome
- **Input Features**: Customer demographics, economic indicators, historical campaign information
- **Objective**: Maximize relevant performance metrics while aligning with business priorities
- **Constraints**: Imbalanced data, mixed feature types, and the need for interpretability

The model should not only perform well statistically, but it must also provide insights that can be translated into **marketing strategies and operational decisions**.

5. Dataset Description

Source

The dataset for this case study comes from the Bank Marketing Dataset available on Kaggle, which is derived from the UCI Bank Marketing Data and reflects responses from direct marketing campaigns conducted by a Portuguese banking institution.

Dataset Characteristics

- **Instances**: 11,000+ customer records
- **Attributes**: ~17 primary features
- **Target Column**: deposit (binary: yes/no)
- **Feature Types**: Mix of numerical and categorical variables
- **Missing Values**: Handled through standard preprocessing or imputation

6. Target Variable

Column	Description
deposit	Subscription outcome ("yes" if customer subscribed to a term deposit, "no" otherwise)

This serves as the **binary target** for the classification model. A positive label (yes) indicates a successful conversion — a customer agreeing to open a term deposit.

7. Sample Feature Groups & Descriptions

Instead of listing all raw columns, the following feature groups describe the **types of information available**, along with representative features and their business relevance:

7.1 Customer Demographics

Feature	Description
age	Age of the customer (numeric)
job	Type of job/profession (categorical)
marital	Marital status (categorical)
education	Education level (categorical)

These features help describe the **customer profile**, which can influence financial decisions.

7.2 Financial Characteristics

Feature	Description
balance	Customer's bank balance (numeric)

default	Indicator of credit default history (categorical)
housing	Whether customer has a housing loan (categorical)
loan	Whether customer has a personal loan (categorical)

These features provide insight into **financial capacity and risk factors** that can affect the likelihood of subscribing to a term deposit.

7.3 Marketing Campaign Information

Feature	Description
contact	Communication type used (telephone/cellular)
month	Month of last contact (categorical)
day_of_week	Day of the week of last contact (categorical)
duration	Duration of last call in seconds (numeric)
campaign	Number of contacts performed during this campaign (numeric)
pdays	Number of days since the client was last contacted (numeric)
previous	Number of previous contacts (numeric)
poutcome	Outcome of the previous marketing campaign (categorical)

These variables capture aspects of **customer interaction and marketing effort**, which are core predictors of campaign success.

8. Key Challenges in Modeling

Class Imbalance

The target class (subscription) is generally imbalanced, requiring careful metric selection and modeling strategies such as:

- ROC-AUC and PR-AUC evaluation
- Threshold tuning
- Sampling strategies (if needed)

Mixed Feature Types

The dataset contains:

- Numerical features
- Categorical features
- Binary indicators

These must be appropriately preprocessed (encoding, scaling, imputation) to build effective models.

9. Objectives & Deliverables

At the end of the case study, learners should be able to:

1. Frame the bank marketing problem as a **classification task**
2. Preprocess and explore the dataset
3. Build baseline and advanced classification models
4. Evaluate models using appropriate metrics
5. Interpret results in a business context
6. Generate actionable insights to improve marketing strategy

10. Expected Business Outcomes

- A predictive model that estimates the likelihood of term deposit subscription
- Improved marketing efficiency by targeting high-probability customers
- Insights into which customer groups are most responsive to campaigns
- A framework that can be adapted to similar marketing or customer conversion problems

11. Real-World Relevance

This case study reflects common real-world requirements in banking and marketing analytics — balancing statistical performance with business value, interpretability, and operational cost considerations. By completing this project, learners gain practical skills directly transferable to industry classification workflows.